

1. Authentication
2. Password hashing & Salt
3. Session
4. JWT
5. OAuth
6. MFA(Multi Factor Authentication)

Authentication

Who are you?

1. Personal Info & messages
2. Medical Data
3. Bank data
4. Company Data
5. Government Data

Password hashing & Salt

Password Hashing

User Table

id	username	password
1	loopkaka	test
2	Sangram	abc
3	Dhiraj	test

Hashing Algo - bcrypt, scrypt, Argon2

id	username	password	
1	loopkaka	ye54432ha	test
2	Sangram	da6734uyga	
3	Dhiraj	ye54432ha	test

Salt

A random string/data add before hashing

id	username	password
1	loopkaka	uyygef78234vdjh
2	Sangram	da6734uyga
3	Dhiraj	b233fuyg873223

Hashing VS Encryption

Hashing:

on-way encryption

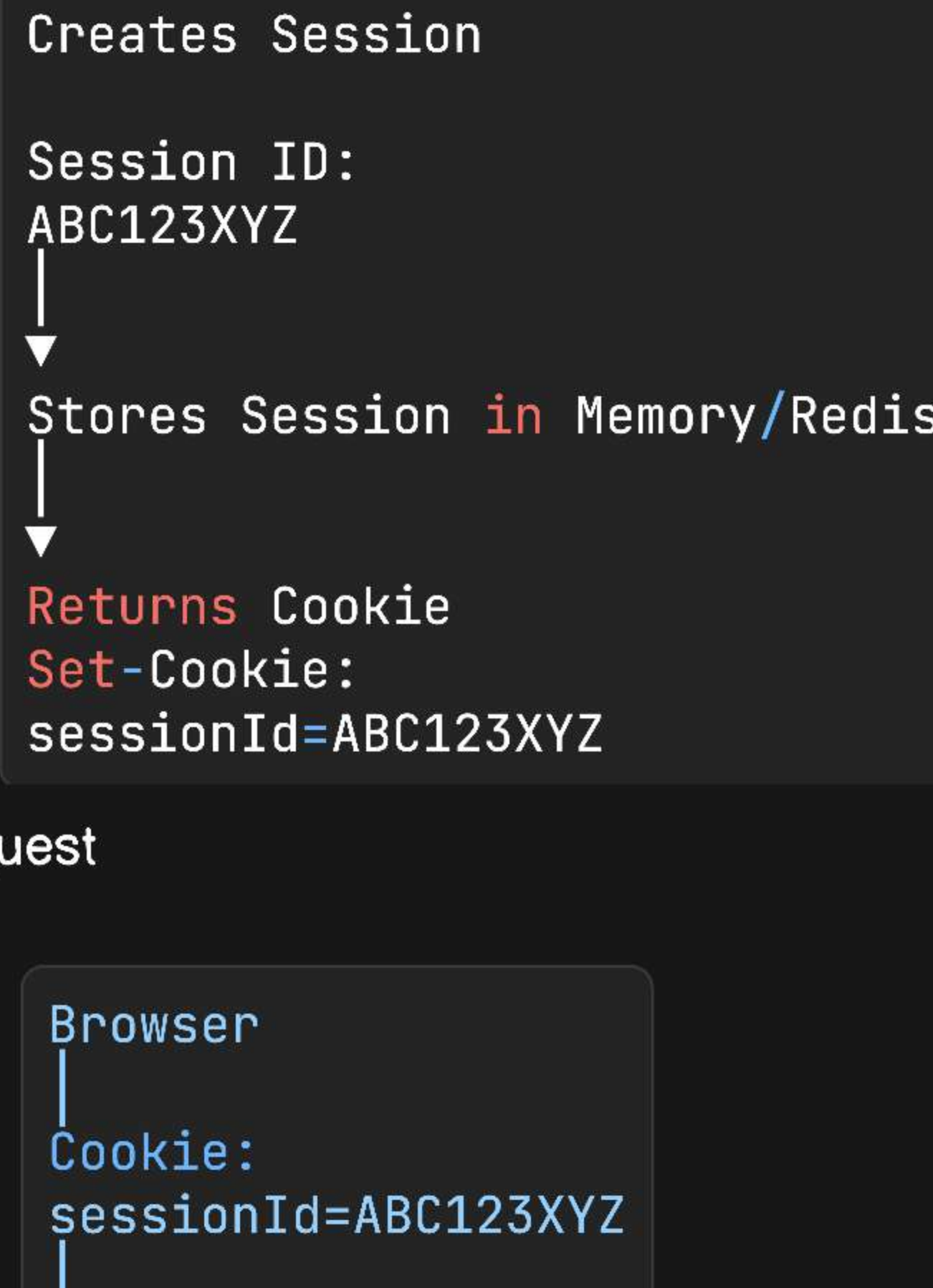
test -> 68732vfvdsvjv

Encryption:

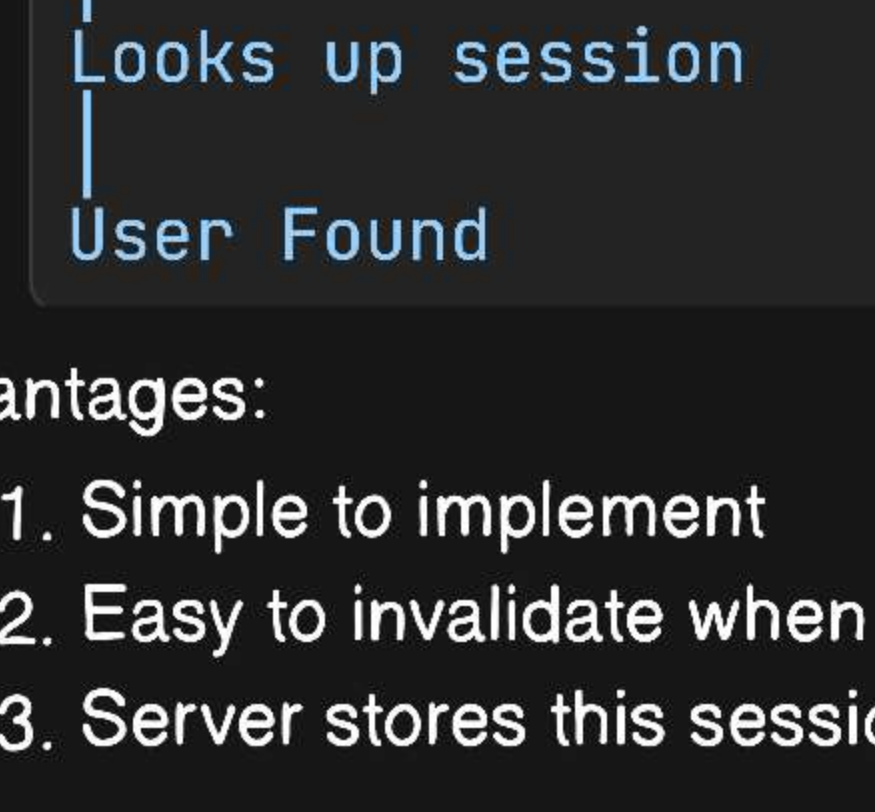
two way encryption

test-> uihdas uyt3432 -> test

Session Authentication



Next Request



Advantages:

1. Simple to implement
2. Easy to invalidate when loggedout
3. Server stores this session id

Disadvantages:

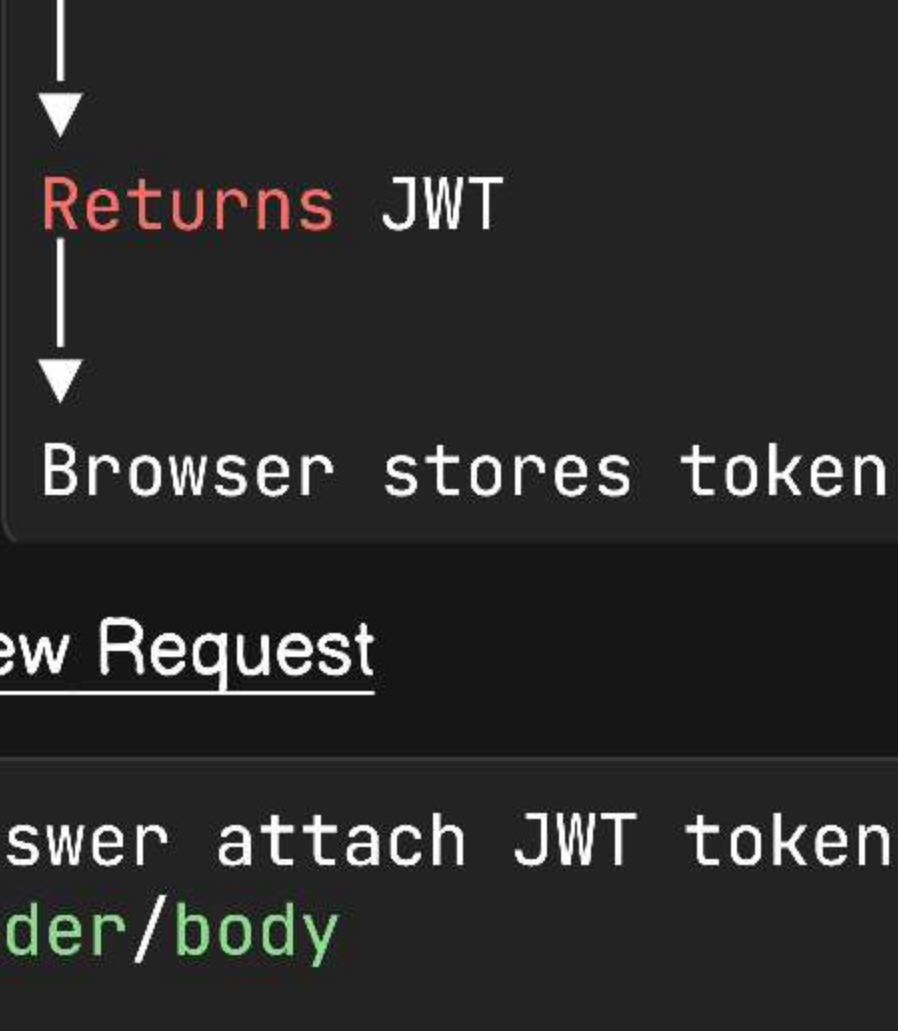
1. Hard to scale
2. Session can be hijacked
3. Session can be temperd

JWT Authentication

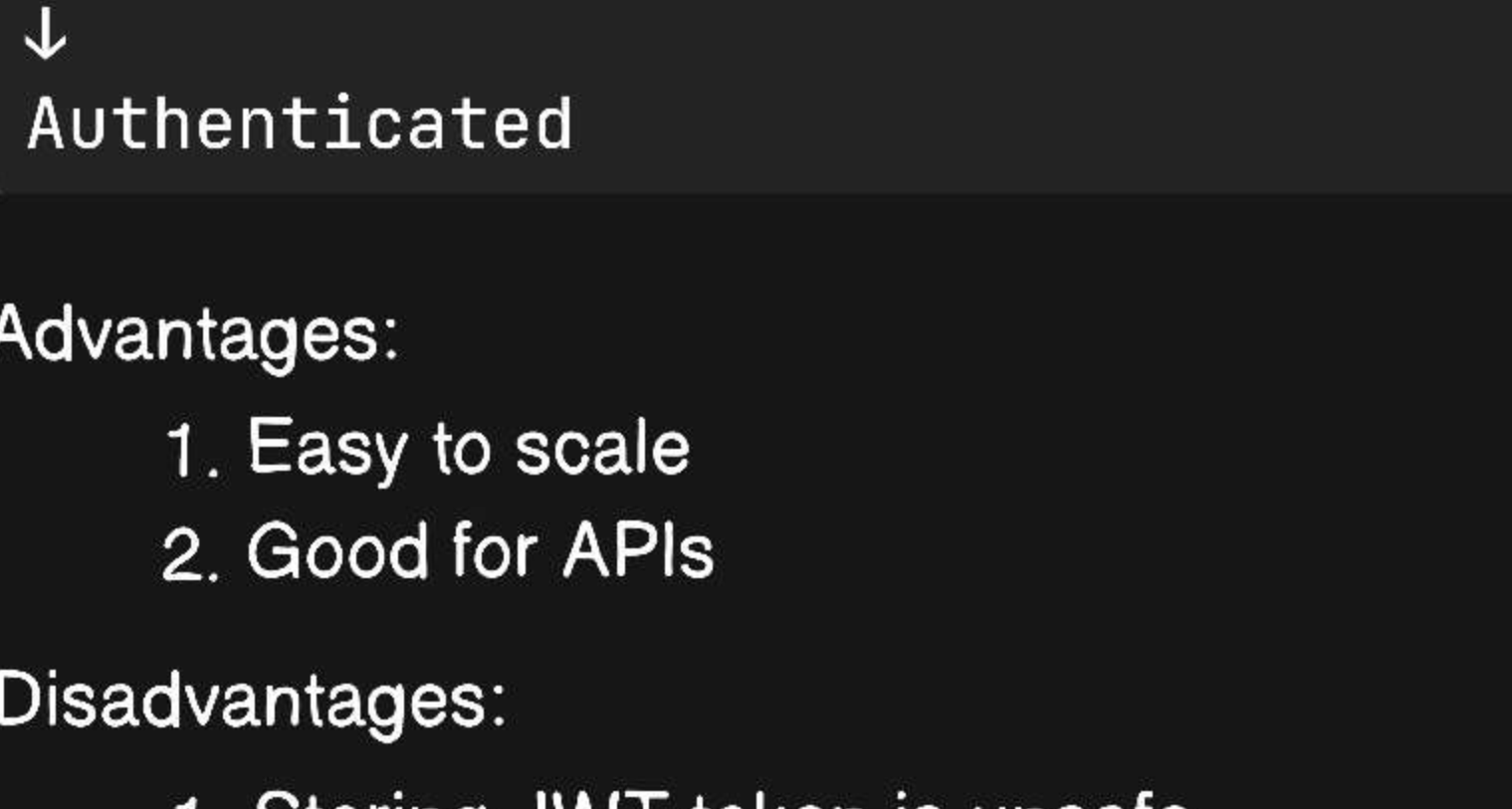
JSON Web Token

xxx.yyyy.zzzzz

1. header - Contains token type and signing algorithm
2. payload - user id, email, role, exp
3. signature - Secret key



New Request



Advantages:

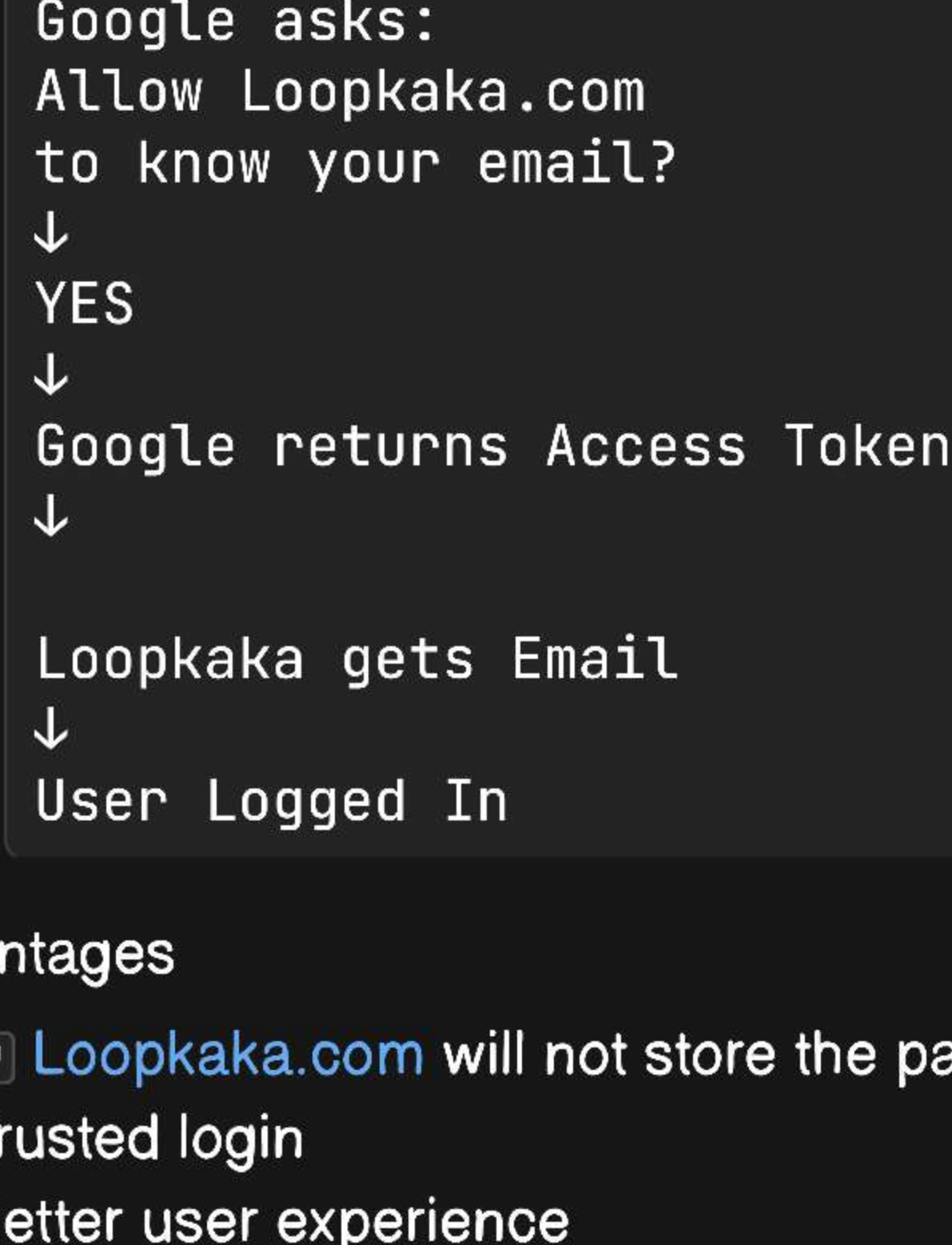
1. Easy to scale
2. Good for APIs

Disadvantages:

1. Storing JWT token is unsafe
2. Token can be stolen
3. Token can be tempered

OAuth

loopkaka.com



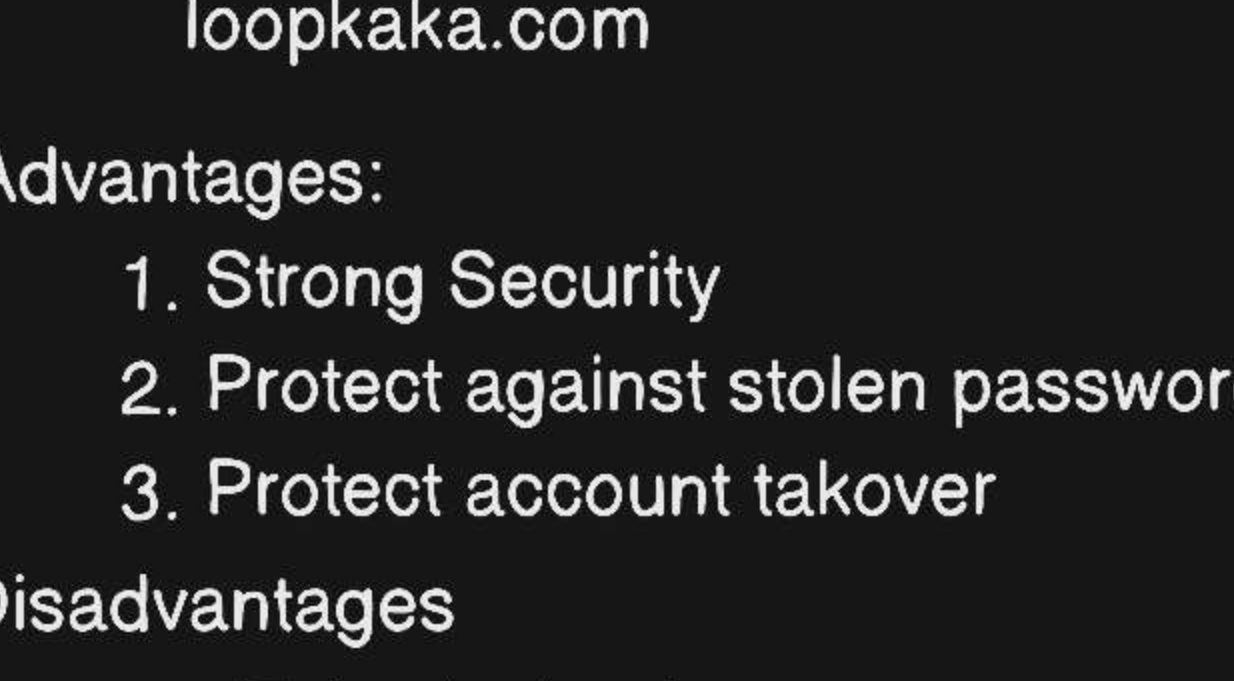
Advantages

1. [Loopkaka.com](#) will not store the password
2. Trusted login
3. Better user experience

Disadvantages:

1. Complex implementaion
2. Dpends on 3rd party provider
3. Provider server may go down

MFA(Multi Factor Authentication)



Advantages:

1. Strong Security
2. Protect against stolen password
3. Protect account takeover

Disadvantages

1. Extra login step
2. User may lose device
3. Account recovery is difficult

